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# Biodiversity and Nutrition in Rice-Based Ecosystems; the Case of Lao PDR

**Caroline Jane Garaway · Chantone Photitay · Khampet Roger · Lieng Khamsivilay · Matthias Halwart**

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**Abstract** This paper presents results from a year-long household survey in which the aquatic resource collection and consumption activities of 240 households across Lao PDR were studied to assess the diversity of species used, their role in household food security and the overall importance of ricefield habitats in this respect. Results show that aquatic biodiversity, under threat in rice-based ecosystems, plays a larger role in household consumption than previous estimates. More than 90 % of these resources are collected by households themselves and the greatest quantities from ricefield habitats. This seasonal aquatic environment is therefore the principal habitat from which households acquire aquatic animals, both to eat fresh and to process and store for use during nutritionally vulnerable times of year. The importance of these habitats therefore goes far beyond their use for rice production and this multi-functionality needs to be understood and addressed in agricultural, conservation and food security policy.

**Keywords** Wild foods · Nutrition · Biodiversity · Aquatic animals · Ricefield ecosystem · Livelihoods · Fisheries

## Introduction

In conservation biology, it is recognised that there is a deficit of information regarding the status of biodiversity in human modified landscapes (Chazdon *et al.* 2009). A study of conservation literature (Fazey *et al.* 2005) showed that very few studies focused on areas under intense human pressure (agricultural landscapes, coastal and urban areas), with most research carried out in relatively intact habitats (such as reserves) with a high degree of plant and animal biodiversity. These areas, however, are not typical of most of the world's landmass and it is estimated that more than half of all species exist principally outside protected areas, mostly in agricultural landscapes (Blann 2006). Nearly one third of terrestrial lands have agricultural crops or planted pastures as the dominant land use (Scherr and McNeeley 2008). At the same time, an estimated 1.1 billion people live in the world's 25 biodiversity hotspots (Myers *et al.* 2002), most of them agriculture-dependant. As noted by Scherr and McNeeley, agricultural lands perform many ecosystem services and provide essential habitat to many species (2008:480).

The study of biodiversity has become part of the food, agriculture and nutrition agenda in as much as wild, gathered species and under-utilized and unexploited food sources play a key role in global food security (Bharucha and Pretty 2010; Penafiel *et al.* 2011). Trends in nutritional research and intervention have moved away from a focus purely on sufficient dietary energy provision to a focus on the importance of micronutrients and the need to adapt nutrition interventions to the diversity of individual and community needs (Burlingame *et al.* 2006). At the same time future population needs will require conserving biodiversity and increasing utilization of the genetic bases of food species (FAO 2004; Toledo and Burlingame 2006; Godfray *et al.* 2010; Frison *et al.* 2011). Wild food species are declining in many agricultural landscapes (MEA 2005; Foley *et al.* 2005) and biodiversity of farmland usually declines with increasing yield (Fazey *et al.* 2005; Green *et al.* 2005; Mattison and Norris 2005).

This paper analyzes and quantifies aquatic biodiversity in rice-field landscapes to livelihoods in Lao PDR. Much of Lao PDR, like other countries in south east Asia, is dominated by ricefield landscapes, with the majority of the

C. J. Garaway (✉)
UCL, London, UK
e-mail: c.garaway@ucl.ac.uk

C. Photitay · K. Roger · L. Khamsivilay
LARReC, Vientiane, Lao PDR

M. Halwart
FAO, Rome, Italy

population engaged, amongst other things, in semi-subsistence agriculture in rural areas. Whilst protein energy malnutrition is in decline in the region (Winichagoon 2006), nutritional problems still exist, brought about in part by the strong annual and seasonal variation in food supply (James 2006). In Lao PDR, Dyg identified the causes of malnutrition as the combination of extreme and/or chronic poverty with a lack of micronutrients such as vitamin A, calcium, iron and iodine (2006). Dietary diversity is seen as a key solution to nutritional problems in the region (Dyg 2006).

Much of this dietary diversity comes from aquatic animal resources as sources of calcium (Hansen *et al.* 1998), vitamin A (Roos *et al.* 2003), and fatty acids (Elvevoll and James 2000). In this specific study area, aquatic species were found to be an excellent source of calcium, iron and zinc (nutrients that are scarce in rural Laotian diets) (Nurhasan *et al.* 2010).

In Lao PDR, and other countries of the Lower Mekong Basin, fish and other aquatic animals are eaten regularly by almost all people, providing a major source of protein and these essential elements (Hortle 2007). Subsistence fishing is carried out by almost everyone who has convenient access to water (Claridge 1996; Garaway 1999, 2005; Nguyen Khoa *et al.* 2005). In a meta review of 20 separate surveys covering four countries in the Lower Mekong Basin, Hortle (2007) estimated per capita consumption of inland fish and Other Aquatic Animals (OAAs) averaged 34 kg/year as actual consumption and 45.5 Kg Fresh Weight Animal Equivalent (FWAE). The country average for Lao PDR was below this at 29 kg/year (43 FWAE) and inland fish and OAAs were estimated to provide approximately 50 % of animal protein.

Quantitative evidence of the use of aquatic resources and their seasonal and geographic variation is rare. Over the past two decades there have been a number of studies on the use and importance of aquatic resources. (Meusch 1996; Garaway 1999, 2005; Morales *et al*. 2006; Nguyen Khoa *et al*. 2005; Noraseng 2001). However these studies have tended to be fairly small-scale and localized, and have concentrated mainly on fish and not OAAs and on total quantities as opposed to species diversity. Data collection of this type is difficult due to the strong geographic and seasonal variation in aquatic resource collection and to the fact that collection is ubiquitous but actual quantities per person are low. Also the products are frequently used for personal consumption or sold locally and are therefore relatively invisible at the national level (Halwart 2008). As a result, national statistics tend to underestimate and therefore undervalue these resources. The precise origin of the fish or OAA catch has also rarely been the principal focus though when it has, it has highlighted the importance of farmer-managed systems (Meusch 1996; Meusch et al. 2003; Gregory and Guttman 2002; Schwartz and Morton 2002; Morales *et al.* 2006). Indeed, in an earlier study to that described in this paper, local people across four countries in the region including Lao PDR were found to utilise 145 fish species, 15 mollusc, 13 reptile, 11 amphibian and 37 plant species from rice-field ecosystems, showing that rice-fields were far more than a source of rice (Halwart and Bartley 2005).

Quantifying the importance of aquatic resources originating from rice-field ecosystems and its seasonal and geographic variation has never been more important. Government undervaluation of these resources combined with the fact that emphasis is on increasing rice yields, the key commodity for local and national food security, means such resources are under threat. Several authors in the region have argued that the abundance of important aquatic animals is threatened by agricultural intensification (Elvevoll and James 2000; Soubry 2001; Beaton 2002; Gregory and Guttman 2002). Well known threats to aquatic biodiversity include poorly managed irrigation or other water related practices that prevent the triggering of natural spawning cycles, act as barriers to migration routes, or cause waterlogging or salinization. Production practices such as the application of pesticides and use of other agrochemicals also may threaten biodiversity (Hazell and Wood 2008).

This paper describes the results of a year-long survey conducted between October 2006 and October 2007 that quantified household collection and use of aquatic resources in 240 households, over three Provinces and 12 districts within the Lao PDR. The study was the result of collaboration between the United Nations Food and Agriculture Organization (FAO) and the Department for Livestock and Fisheries (DLF), Lao PDR. Despite their known importance, quantifying the use of aquatic plants was beyond the scope of this survey and they are therefore not included, though clearly worthy of further investigation.

## Materials and Methods

The need for quantitative data with maximum geographical coverage favoured a questionnaire- based approach over other more in-depth qualitative methods. Logistic constraints set a limit on the number of households that could be monitored and these were chosen to represent the broad geographical range of conditions in Lao PDR, as discussed below.

### Site Selection

Firstly, three provinces (Champassak (lowland floodplain), Savannakhet (lowland floodplain and upland areas) and Xieng Khouang (upland/mountainous)) were chosen as being broadly representative of the different agro-ecosystem and topographical zones in the country and 80 households were monitored from each. Within each province, four districts were selected that were broadly characteristic of the province but varied with respect to distance from the provincial capital while being accessible by road throughout the year.

With regards to selection of villages and households, previous studies have shown that inter-village variation in (fish) catches was far greater than intra-village variation (Garaway 2005) and that this was probably linked to availability of aquatic habitats, both permanent and seasonal (Garaway 1999). It was therefore decided that the number of villages covered would be maximised resulting in five households being chosen in each of the four villages. Thus a total of 48 villages and 240 households were sampled. Villages were selected on the basis of their distance from a central market and availability of aquatic habitats, both factors believed to affect aquatic resource collection activities. Households (here defined as those people who 'contributed to' and ate from the 'same pot') were selected randomly.

### Household Questionnaire on the Acquisition and use of Fish and OAAs

The survey was designed to measure the quantities of fish and OAAs coming into a household from different sources (bought,

**Table 1** Methods for estimating weight of fish and OAA's

| Main types of animal and forms in which acquired | Common units of measurement/method of description | Method of weight estimation |
|---|---|---|
| Small fish (5 cms) | Noodle bowls [1] | Market samples [2] |
| Shrimp | Noodle bowls | Market samples |
| Large fish (>5 cms) | 'Fish sticks' [3a] | Length/weight relationships[3b] |
| Parts of large fish (bought) | Price | Price/Kg |
| Dried fish (bought) | Price | Price/Kg converted to FWAE using conversion factors from Hortle 2007 |
| Fermented fish (*ba dek* [4]) | Noodle bowls | Market samples and conversion factors from previous research [5] |
| Small frogs | Noodle bowls | Market samples |
| Large frogs | Count [6] | Market samples |
| Small and medium crickets and grasshoppers | Noodle bowls | Market samples |
| Other insects | Count or bowl as appropriate | Market samples |
| Small and medium sized crabs (caught) | Noodle bowls | Market samples (includes shells) |
| Medium sized crabs (bought) | On sticks (count number of sticks) | Market samples (includes shells) |
| Small and medium sized snails | Noodle bowls | Market samples (includes shells) |
| Tadpoles | Noodle bowls | Market samples |
| Snakes | Fish sticks/ruler | Group 1 fish species [3b] |
| Other animals | Count or bowls as appropriate | Market samples |
| | If bought (Price) | Price/Kg |

[1] Noodle bowls were a basic item of household crockery. Four sizes were commonly recognised and used. Bowl A (250 ml); Bowl B (400 ml); Bowl C (700 ml); Bowl D (1,200 ml). These 4 different bowls (marked, A, B, C, D) were used as visual aids when conducting the survey

[2] Market samples. Multiple samples were bought from markets and weights (in different bowl sizes) measured by the project team This was repeated several times and the average recorded weight chosen

[3] It was known from previous research (Garaway 1999) that, generally, fishermen were not able to adequately estimate weights of fish they had caught, but that they and fish buyers could accurately estimate, via recall, the length of fish to within 5 cm in a 5–25 cm range. This range covered 95 % of fish caught. Villagers' estimates of the lengths of their fish (using different length sticks (termed fish sticks) were used to estimate weights using length/weight relationships. Firstly, length/weight relationships were calculated for some of the most common species of fish in these Provinces using the following relationship: Weight = a.lengthb, where a is constant and b ≅ 3. Data for these calculations came from earlier research (Garaway 1999) and other literature sources (mainly Fishbase 2008). Next, on the basis of similarity of coefficients, the common fish species were separated into three groups: 1. *Anguila* spp, *Crossocheilus siamensis*, *Xenentodon cancila*, *Mastacembelus armatus*, *Macrognathus* spp., *Monopterus albus* (Fluta alba); *Mastacembelus favus*. 2. *Cyprinus carpio* and *Oreochromis niloticus*. 3 All other species. The coefficients were recalculated so that they all referred to an exponent of three and the average weight for each group of fish for differing lengths was calculated. Lastly, "fish sticks" were designed to provide an appropriate tool for using the length weight information in the field. These were chopsticks of the 5 different lengths (5 cms, 10 cms, 15 cms, 20 cms, 25 cms), with the corresponding weights for each group of fish directly marked on them to allow ease of use in the field

[4] Common form of processed fish, frequently used daily. It is made from small fish and shrimp, combined with rice-bran, salt and water, and is the most common way of utilising and storing surplus fish caught at relatively abundant times of year. Different grades of *padek* are commonly recognised and figures here relate to average quality (Fishbase 2008)

[5] Estimates of the quantity of fish (FWAE) in average grade *padek* were taken from previous research (Garaway 1999; Hortle 2007), using a conversion factor of 0.75 (includes both the processing conversion factor and preservation factor)

[6] I.e. the number of individual animals

caught, received as gift, exchanged) in the preceding 24 h period, and what it was subsequently used for. Additional information was recorded on fish and OAA collection activities (location of the catch and who caught it) and, where possible, individual species were recorded. The questionnaire instrument was field tested several times by DLF and further improvements were made after the first 6 months of data collection.

To better understand the contribution of fish and OAAs to diet, additional information on other products consumed at mealtimes over the previous 24 h was also collected, though not in as great detail as for the fish and OAAs. Finally, information was collected on household members (age and sex) and their presence/absence at mealtimes in the previous 24 h. This, along with information about the presence of any additional non-household members at mealtimes allowed for the analysis of consumption per Adult Equivalent Units (AEU).

Given the large sample size and the lack of measuring equipment within the majority of Lao households, despite its limitations a 24 h recall method, commonly used in consumption studies, was chosen over direct measurement. In a comparison of interview and measurement methods, Garrison *et al.* (2006) found that people consistently over-estimated consumption of fish and OAAs during interviews, a result that supported findings of earlier research (Garaway 1999). To help overcome these potential biases, several features were built into the design of this survey.

Firstly, people were not asked to estimate the weight of the fish or OAA acquired directly, but were asked to do so indirectly with the use of visual aids (such as number of noodle bowls or jars of a fixed and recognised size) or other proxies, as was most appropriate. This was the case for both fresh and processed aquatic animals, and in the latter case conversion factors were applied to establish fresh weight animal equivalents (FWAEs) (Table 1).

Secondly, because the weight of the fish or OAAs was established at the point they came into the household, it did not have to be estimated again. Instead people were only asked to estimate the percentage of fish being used for the different activities (consumed, stored, sold, exchanged, given as a gift) and these percentages were then converted to weight. This had a significant advantage when looking at household consumption as it is far easier to estimate the weight of a product whole

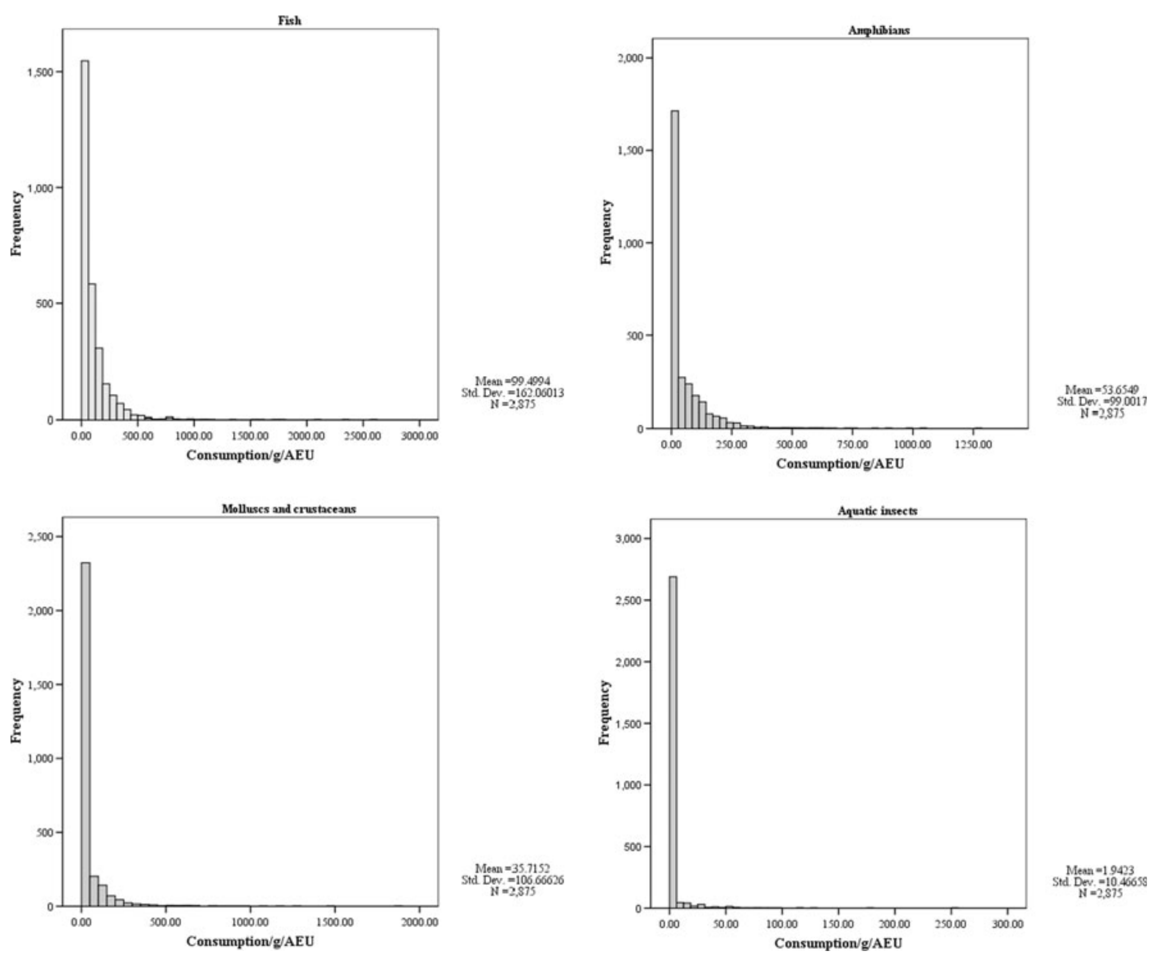
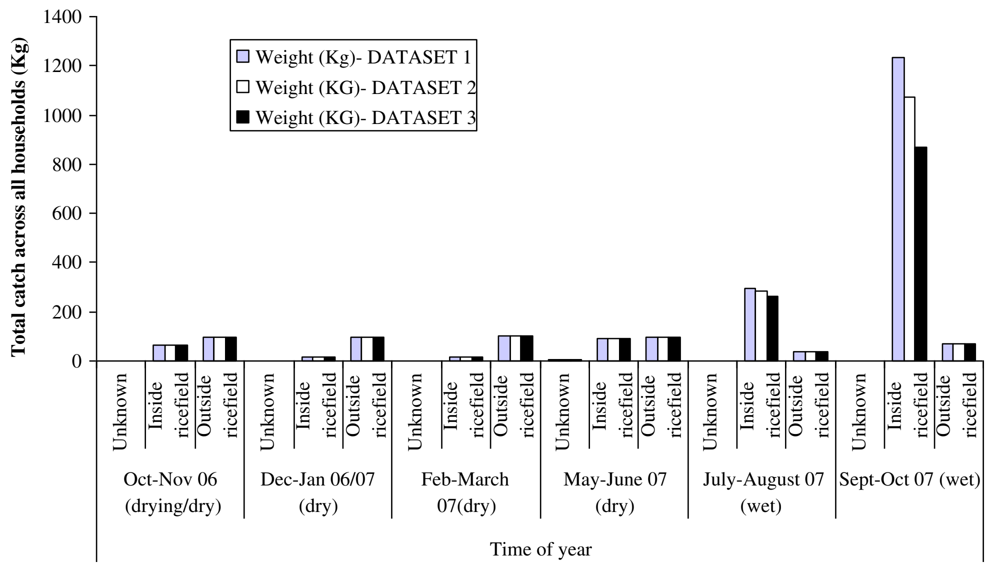



**Fig. 1** Histograms showing consumption of fish and OAA's (g/AEU/day)

**Fig. 2** Seasonal variation in weight and origin of fish catch (Includes the species *Macrobrachium lanchesteri*) (Separated into DATASETS 1–3 (See "Materials and Methods"))




and on its own than when it has been in some way combined with other ingredients. This issue is a likely explanation of the overestimates described by Garrison *et al.* (2006), and also the reported disparities between catch and consumption values found in fisheries surveys where the quantities of fish consumed is frequently inexplicably higher than that being caught (Singhanouvong and Phouthavongs 2003)

## Data Collection

The survey was conducted over a 13-month period between October 2006 and October 2007. District DLF staff were responsible for collecting data, with each household being visited 12 times over this period at approximately monthly intervals. Data collection activities were synchronised across all Provinces and Districts, with collection in each round taking no more than 2 weeks. Due to the fact that changing hydrological conditions could greatly affect the availability of aquatic resources, particularly at certain points of the year (such as the draw-down in September–October at the end of the wet season) hydrological data (specifically, the level of water in households' rice-fields) were also collected.

## Data Preparation for Main Analysis

The data were entered into an Access database by central DLF staff, checked rigorously for error, and then analysed in SPSS.

### *Missing Values*

For analysis, missing values (approximately 20 % of all records in first 6 months and <1 % in second 6 months) were replaced by the series' medians for that monitoring period, split by Province, animal type and method of acquisition. The series medians were used as opposed to the arithmetic mean to minimise the effect of some extreme top end outliers.

### *Outliers*

Extreme positive outliers were most notable with fish catches. In particular, a small number of very high fish catches (caught in the drawdown period Sept/Oct 07) were heavily influencing mean values over the whole data set. To deal with this, the median was generally preferred over the mean when comparing averages.

However when comparing total quantities (e.g., percentage of fish used for different purposes or coming from different habitats), the impact of these extreme high end outliers had to be considered directly. Regarding the accuracy of these data, it was expected that very large catches might occur in the September/October period. However, catches of this magnitude (10 times bigger than the median value) were not expected. That being said, there are no comparable data from other studies, and with no information on the gears used or total fishing effort, it is very difficult to reject these results as inaccurate with any confidence. As a result, three different datasets were created dealing with the extreme cases in different ways,[1] and, where relevant, they are presented together.

[1] DATASET 1 : Where all data was presumed to be correct and included, as entered; DATASET 2: Where any individual catches greater than 20 Kg (5 cases) were cut back to 20 Kg only; DATASET 3: Where all values greater than 20 Kg (5 cases) are seen as mistakes and removed from the dataset.

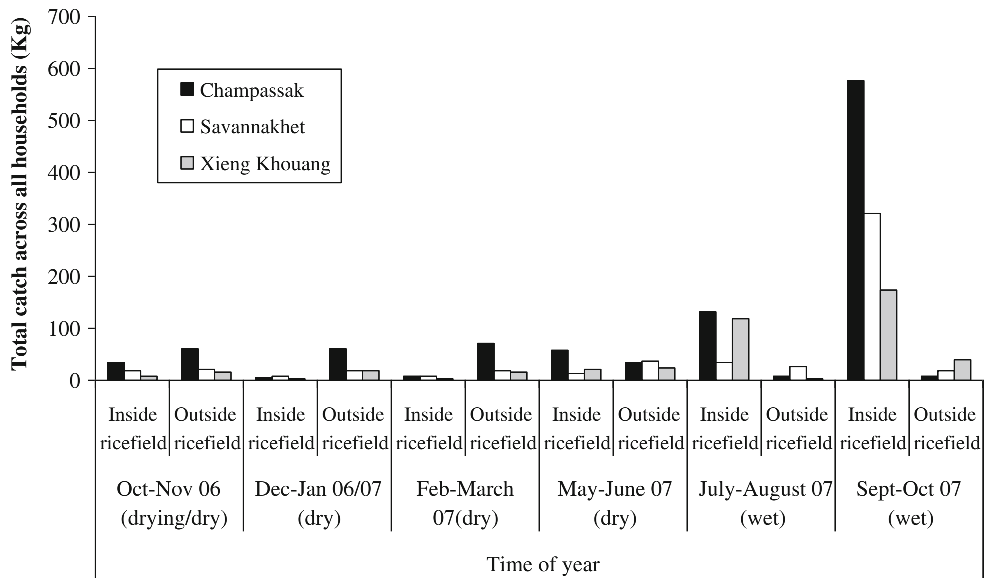



**Fig. 3** Provincial variation in fish catch (Given that the different datasets (see Fig. 2) produce a very similar pattern, results presented here are for Dataset 2 only)

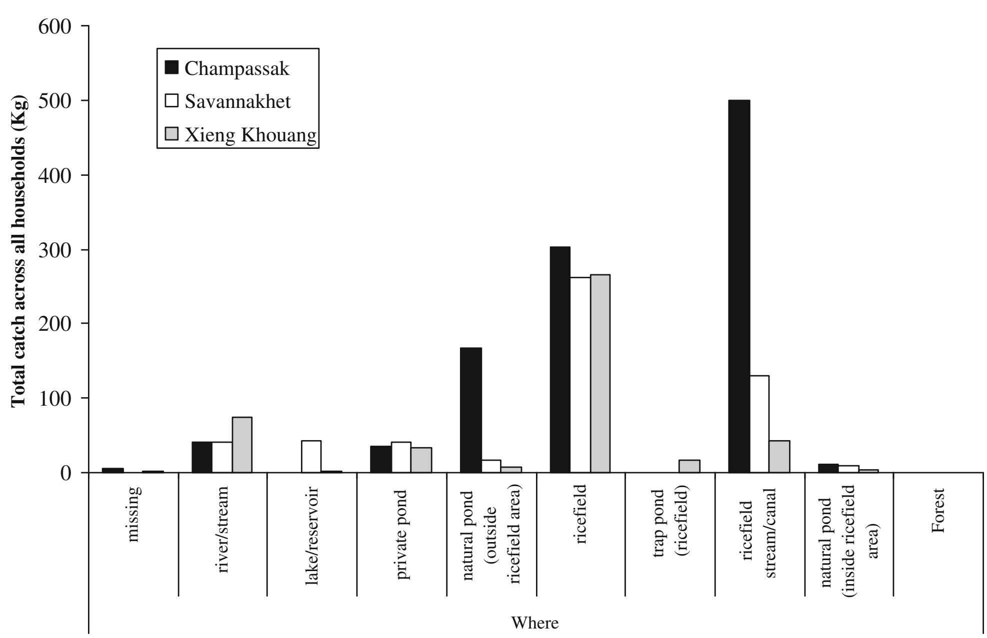
*Calculation of Adult Equivalent Units (AEUs) for Estimating Consumption of Fish and OAAs*

Adult Equivalent Units (AEU) were calculated using the energy requirements of 2005 for the Lao PDR taking urban/rural location and subsequent estimates of levels of physical activity into account. Both family members who were present for meals and any visitors on that day were included in the calculations. AEU was normally distributed (Mean = 4.6, SD=1.7, $N$=2880). At a Provincial level, mean AEU was 4.3 (Champassak), 4.7 (Xieng Khouang) and 4.8 (Savannakhet).

*Methodology for Analysing Consumption Data*

As can be seen from the individual histograms (Fig. 1), all of the distributions are highly skewed, have a large number of zero values and also some extreme high end outliers. Given this, several issues needed to be considered when estimating *total annual consumption*. Again the median was considered a better measure of central tendency than the arithmetic mean. However, large numbers of zero values can represent problems when using the median, for when consumption of a particular

**Fig. 4** Origins of fish catch (Kg)



animal group is infrequent (<50 % of instances), the median is actually zero.

To deal with this, the two events–'eating' and 'not eating' the animal group concerned–were treated differently for analytical purposes and a two-step process was taken. Firstly, instances of 'eating' versus 'not eating' were directly compared by establishing the *frequency* with which different products were consumed (and how this varied with Province and time of year). Secondly, in those cases where the product had been eaten, the average *quantities* consumed (on a g/AEU basis) were investigated, using the median as the measure of central tendency. When combined (frequency x average (median) value), this gave an estimate of total consumption and its variation, that controlled for the particularities of this dataset and allowed a more meaningful analysis into what was actually happening at the 'average' household level.

## Results

### Aquatic Animal Diversity

Table 2 shows the range of species, or families of species, that could be identified by respondents in their local language and subsequently ascribed scientific names by the central government staff. Lack of respondent knowledge and/or an inability to ascribe a scientific name means that this list is by no means definitive, with many animals grouped as 'unidentified species.' However, the list serves as a useful starting point to show the minimal range of aquatic biodiversity utilised by households and frequency of collection.

There were a total of 44 fish species recorded, *Channa striata* and *Clarias batrachus* being the most common. Interestingly, the exotic *Oreochromis niloticus* was also commonly found, being fifth on the list. Whilst some of this catch came from private ponds, the majority did not.

Only three species of amphibian were identified by scientific name, with most of the sample remaining unidentified. In the Oct 2006–March 2007 period, a large number were a small frog frequently referred to in Lao as 'kied na' (rice-field frog').

With regards to other animal types, much of the sample was unidentified, with only four named mollusc and reptile species, two crustaceans, and no aquatic insects that could be identified by their scientific name. Aquatic biodiversity then is higher in the region than can be reported here (as has been reported elsewhere, e.g., Halwart and Bartley 2005), but even these results show that a huge range of aquatic species are collected and are contributing to household livelihoods.

### The Importance of Rice-Field Habitats

#### *Fish*

Habitats are categorised as being inside the rice-fields (rice-fields, rice-field streams and canals; trap ponds; small natural ponds or swamps contained within the rice-field area) or being outside (lakes/reservoirs; natural ponds; private ponds; rivers and streams; forest) (Fig. 2). The percentage of fish catch coming from inside versus outside rice-field areas varies quite considerably with the time of year, being at its lowest from the rice-fields, after a steady 6 month decline, in the dry Feb/March period (<5 % of total). Moving into the wet season, by the May/June period, it starts to increase but is still less than 50 %. By July however, rice-field habitats

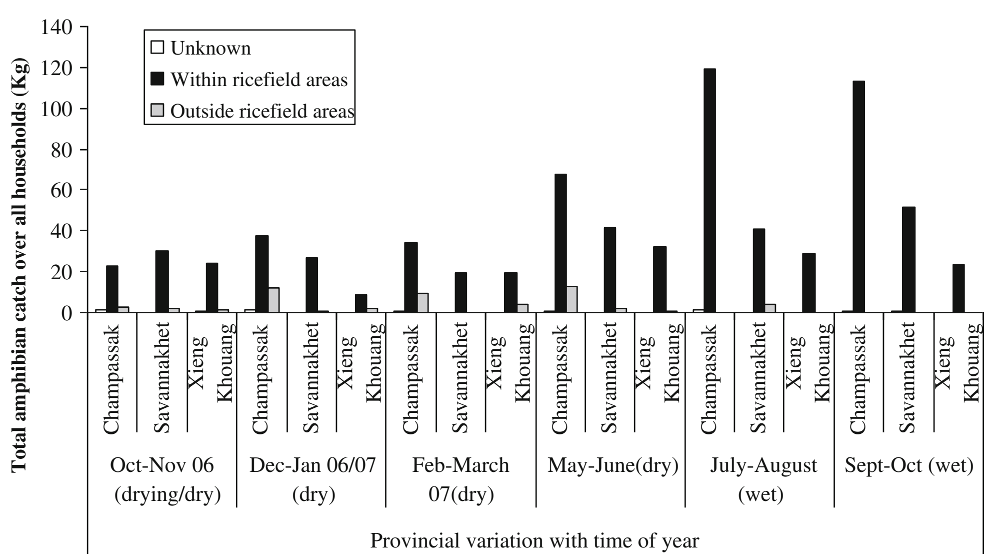



**Fig. 5** Provincial and seasonal variation in amphibian harvest

contribute more than 90 % of total fish catch. This pattern holds whichever dataset is used. Turning to quantities, catches are generally much higher in the July–October period (rainy season) and therefore, on an annual basis, areas within the rice-fields are a far more important contributor to annual fish catch (75 %) than areas outside (25 %).

This general picture varies in different parts of the country and differences are quite large (Fig. 3). In the lowland alluvial plains of Champassak, rich in water resources, substantially more fish is caught than in the other two Provinces where water resources are less plentiful. However, with respect to the importance of habitats within the rice-fields, all Provinces show much the same pattern described above (with some variation in the transition period between the dry and wet seasons (May–June)).

Looking more specifically at habitats (Fig. 4), in Champassak Province the rice-field streams are the most important followed by the rice-fields themselves, whereas in the other two Provinces the rice-fields contribute the most. The other significant sources of fish in Champassak are natural ponds and swamps outside of the ricefield areas. In the mixed lowland/upland areas of Savannakhet the other main sources of fish include natural lakes, reservoirs and rivers, whilst in the upland areas of Xieng Khouang, rivers are the only other significant contributor aside from private ponds, which make a similar contribution in all Provinces.

### *Amphibians*

Habitats within the rice-fields account for more than 90 % of amphibian species, a pattern that holds in all the different agro-ecological zones (Fig. 5). From May onwards the rice-fields themselves are the most important habitat and from July to October catches are almost exclusively from here. Other habitats include natural swamps inside and outside the rice-field areas and the rice-field streams and canals. As for fish, amphibian catches are far higher in the lowland floodplains of Champassak than elsewhere.

### *Molluscs*

For molluscs, habitats in the rice-field areas are most important in the rainy season and overall contribute 63 % of total catch (Fig. 6). Again total catches are far higher in Champassak than the other Provinces and it is the only Province with a significant dry season catch (predominantly, and unsurprisingly in areas outside of the rice-fields).

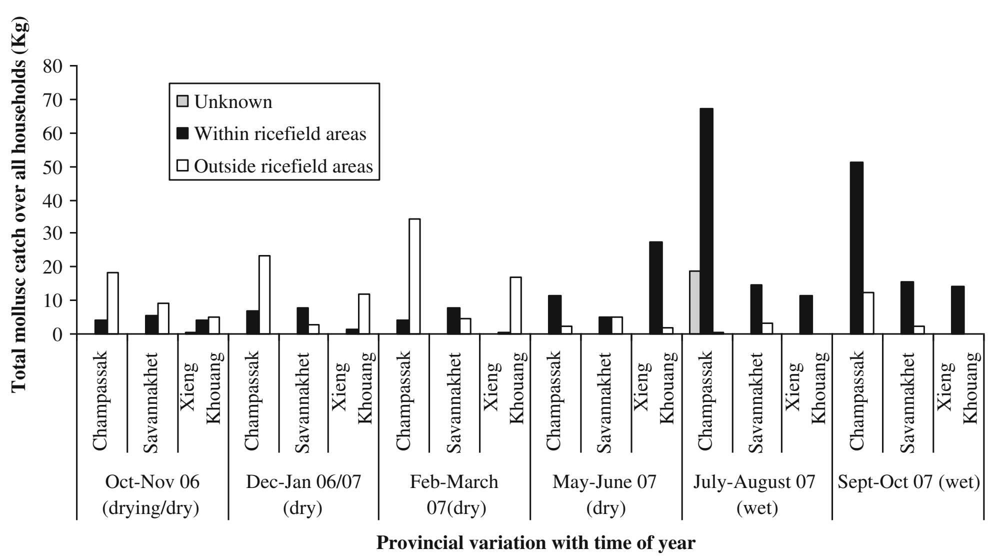
### *Crustaceans*

In accordance with the OAAs, habitats in the rice-field areas play by far the greatest role throughout the year (94 % of total catch), and completely dominate from May to October (Fig. 7).

### *Aquatic Insects*

Finally, in contrast to the aquatic animals discussed thus far, for aquatic insects habitats outside the rice-field areas are as important as those within, more so in some periods (Fig. 8). In general the pattern of catches is very different to fish and the OAAs. Catches peak in the middle of the year and are, in fact, very low towards the end of the wet season when catches for everything else are much increased. With overall weights so low, though aquatic insects are recognised as an important protein-rich source of food our relatively little data do not warrant further discussion here.

**Fig. 6** Provincial and seasonal variation in mollusc harvest



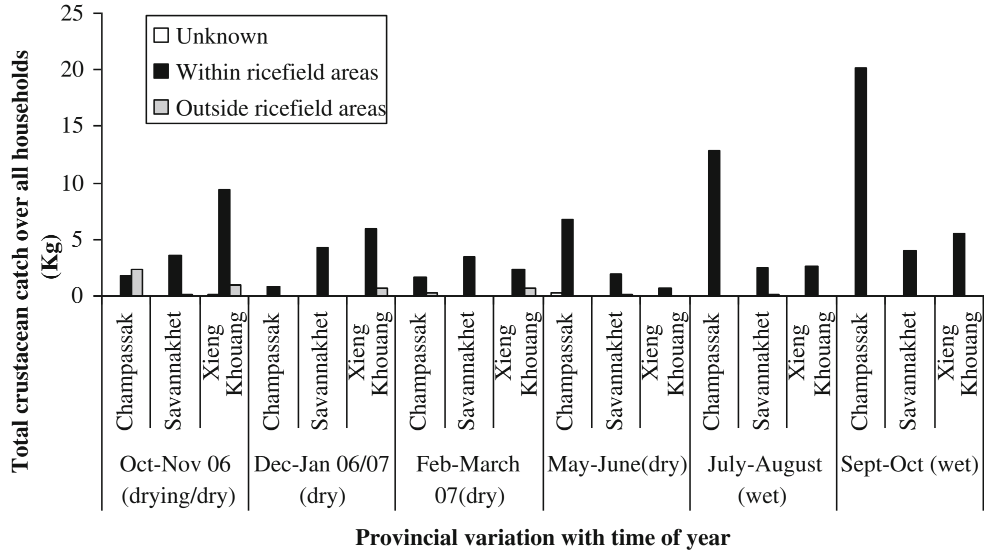



**Fig. 7** Provincial and seasonal variation in crustacean harvest

## Importance of Gathering Activities

Collection of aquatic wild foods is by far the most important means by which they come into the household (at least 90 % of the total) (Fig. 9). The majority of the balance is bought. This highlights the critical role that wild food collection plays in livelihoods across the country and suggests significant household losses if the wild food availability were to decline.

## Utilisation of Collected Aquatic Animals

Across the Provinces the majority of fish and OAAs are either eaten directly or processed and stored. Sale or use as gifts or exchange are far less important (Fig. 10). Direct consumption is lowest for fish (33 %–58 %), though there is considerable Provincial variation, with consumption less and processing and storing[2] higher in Champassak, unsurprising in light of its rich aquatic resources and higher catches. Even so, households still consume significantly more fresh fish on a daily basis than in the other Provinces (see below).

The vast majority of other aquatic animals are eaten fresh: amphibians, molluscs, crustaceans 70–75 %; aquatic insects 90 %. The remaining 25/30 % is split between the other main uses though selling is only significant in the case of amphibians.

Data on consumption of the processed and stored aquatic animals from the household store were collected for the wet season but unfortunately not adequately for the dry season. It is expected that much is used for the household's own consumption, being particularly important in the dry season when it is likely to be the most significant source of animal protein, but it is also probable that some is given away (to friends/family in or outside the area) or sold. Therefore estimates for household consumption that follow are likely to be minimum estimates only.

[2] Fish is most commonly salted and fermented ('Padek') and can be stored at ambient temperatures for extended periods.

## Household Consumption of Fish and Other OAAs

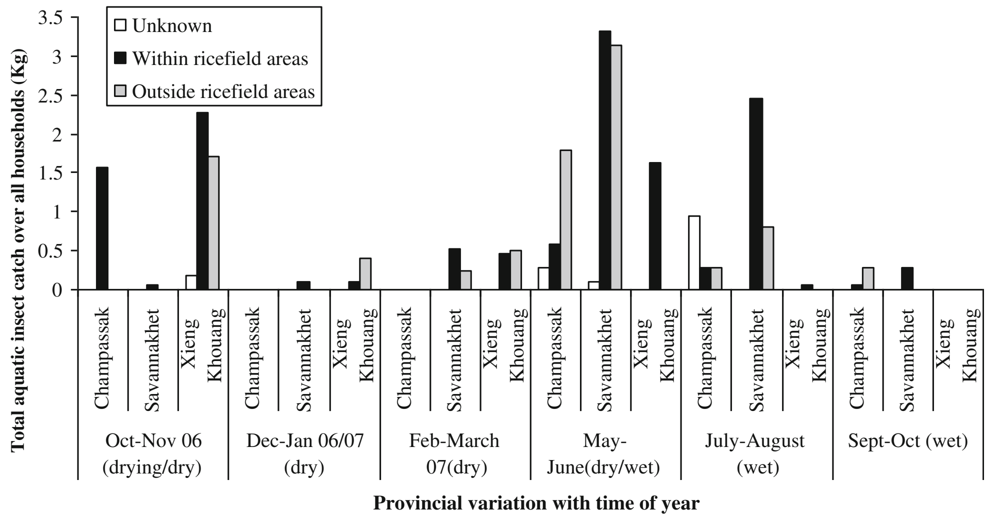
This section focuses on consumption and *all* the fish and OAAs coming into the household, which, as we have seen, are mostly caught by households themselves.

### *Total Annual Consumption of Fish and OAAs*

For all aquatic animals except fish (fresh or processed), the frequency of consumption was less than in 50 % of recorded instances, illustrating why a two-stage approach to estimating total consumption had to be employed (Table 3: the last column combines the data and provides an estimate of total consumption (g/AEU/yr)).

Taken over the whole sample, annual estimates (FWAEs) for consumption are 25.2 Kg/AEU (fresh fish), 15.7 Kg/AEU (amphibians), 9.54 Kg/AEU (molluscs and crustaceans) and .516 Kg/AEU (aquatic insects). The total of 51 Kg is substantially higher than the value for Lao PDR estimated by Hortle (2007) at 43 Kg/capita/yr though results are not directly comparable given that they provide per capita not per adult equivalent estimates; nor is it clear whether exactly the same animals are included. Nevertheless, the difference lies not in the estimates for fish consumption (higher in the Hortle study) but in the estimates for the consumption of the other aquatic animals that are as much as three times higher than reported elsewhere.

**Fig. 8** Provincial and seasonal variation in aquatic insect harvest




### *Geographical and Seasonal Variation*

Using the data presented in Table 3, we calculated seasonal and geographical variation in daily consumption (Fig. 11). While these figures reflect estimated *daily* amounts, in fact the products are eaten less frequently and consumed in greater quantities. However, the figures are useful in showing general trends over longer time periods.

People in Champassak consume considerably more than the other two Provinces, which are more comparable to each other. Fish is eaten more frequently than the other two Provinces (81 % vs. 70 %[3]) and in significantly[4] larger quantities (Mdn = 117 g; IQR = 61 g–209 g vs. Mdn = 73 g; IQR = 37 g–138 g) but frequency of fish consumption is high in all Provinces. Seasonality of consumption is far less marked in Champassak since households have access to more perennial water resources and therefore their fish consumption is more consistent throughout the year. In contrast, elsewhere there is a large increase in consumption in the wet season when the rice-fields become inundated and availability of aquatic resources increases.

Households in Champassak also eat amphibians more frequently than the other two Provinces (57 % vs. 37 %[5]). As with fish, they also eat more per meal accounting for the more than double annual consumption estimates (Mdn = 129; IQR = 75–205 vs. Mdn = 69; IQR = 43–112[6]). Seasonality of consumption is generally similar in all Provinces and far less marked than for fish

Finally, consumption of molluscs and crustaceans is far less frequent than for the other animals. However, they were still eaten slightly but significantly[7] more frequently in Champassak (28 % vs. 22 %). The main reason for higher consumption in Champassak is the significantly[8] larger differences in the quantities consumed per meal (Mdn = 137; IQR = 79–259 vs. Mdn = 84; IQR = 45–130). Again Champassak shows a very different pattern with regards the seasonality of consumption than the other two Provinces.

### Importance of Aquatic Animals Versus Other Foodstuffs

We calculated the percentage of days on which different foodstuffs were eaten over the 6 days monitored between May and October 2007, i.e., the *wet season* only (Fig. 12). If a household ate a product type (disaggregated by individual animal on the questionnaire) in the previous 24 h period, we counted a 'yes' for that foodstuff type, irrespective of how many times it (or another within the same food grouping) had been eaten. The graph therefore represents the presence or absence of a particular food group on any given day, and not the frequency of consumption or the variety present within that food group.

As expected, rice was the most frequently consumed foodstuff (just under 95 %) followed by vegetables (89 %). For all 240 households fresh fish was the most frequent source of animal protein (77 %), with higher levels than other animal types (58 %) and processed fish (70 %) (although the figures for processed fish are heavily affected by extremely low values in Xieng Khouang Province). It should also be remembered that the time of year these data were collected relatively low processed fish consumption is expected given the higher availability of fresh fish (Fig. 12). Next in frequency of consumption are the other aquatic animals (56 %) and oil (20 %). Fish and OAAs feature more significantly

[3] $\chi^{2(1)}=46$, $p<.001$

[4] U=383236; $p<.001$

[5] $\chi^{2(1)}=93$, $p<.001$

[6] $U=112628$; $p<.001$

[7] $\chi^{2(1)}=10$, $p<.001$

[8] $U=35357$; $p<.001$

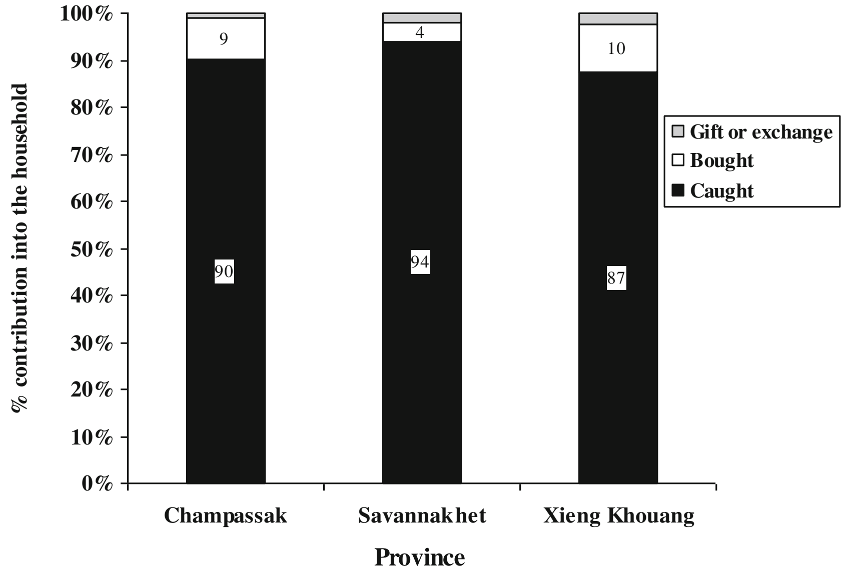



**Fig. 9** Percentage contribution of different activities for annual household fish supply

in Champassak than elsewhere, with processed fish and fresh fish recorded more frequently (almost as often as vegetables), closely followed by the other aquatic animals. In the other Provinces consumption of non-aquatic animals is higher than the 'other aquatic animals' and is closer to fish than in Champassak.

## Discussion

These results demonstrate that a diverse range of aquatic resources play a vital role in household nutrition, that the vast majority of these aquatic resources are collected by the households themselves, and that households are collecting the greatest percentage of these goods from habitats within the rice-field areas, including small streams, ponds and the rice-fields themselves.

Firstly, at an average of 51 Kg/AEU/yr (FWAE), annual consumption of fish and other aquatic animals is even higher than recent estimates (e.g., Hortle 2007 (43 Kg/capita/yr) (FWAE)) and significantly higher than estimates currently used to inform government policy. Much of this difference can be attributed to significantly higher consumption figures for amphibians, molluscs and crustaceans. It should also be remembered that these results for fish consumption are considered minimum estimates as the figures do not include fermented fish (padek) consumption during the dry season. Provincial averages ranged from a substantial 36.3 Kg/AEU/yr (FWAE) to 76.9 Kg/AEU/yr (FWAE) showing these

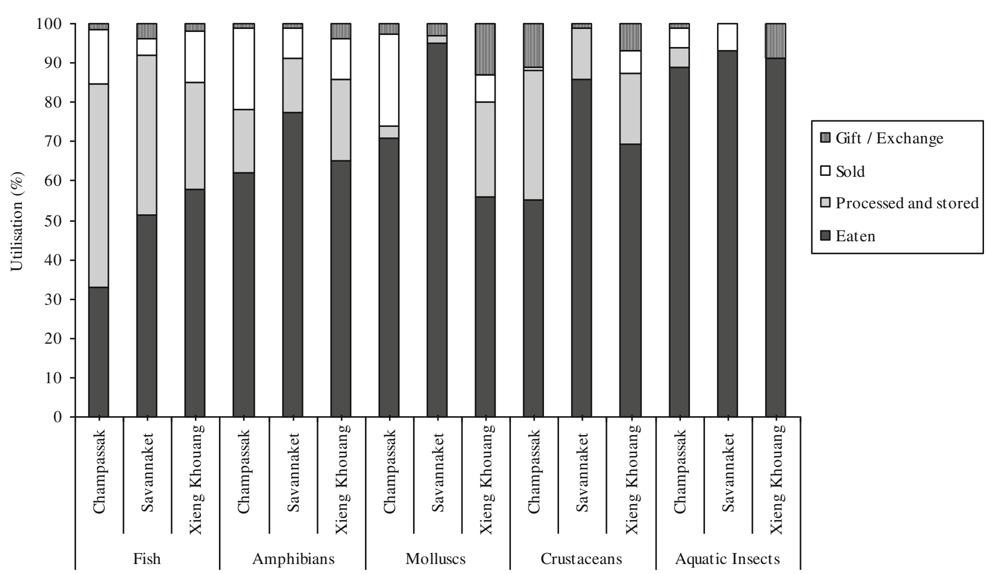



**Fig. 10** Household utilisation of aquatic animal catch

**Table 2** Biodiversity of fish and OAA's

| Animal | Scientific name (where known) | Total times caught | Unknown | Within ricefield Areas | Outside ricefield areas |
|---|---|---|---|---|---|
| Fish | Channa striata (Ophiocephalus striatus) | 689 | 3 | 418 | 268 |
| | Clarias Batrachus | 580 | 3 | 375 | 202 |
| | Fish (species unidentified) | 350 | 2 | 242 | 106 |
| | Anabas testudineus | 279 | 3 | 186 | 90 |
| | Oreochromis niloticus | 242 | 2 | 116 | 124 |
| | Puntius jacobusboehlkei | 217 | 1 | 125 | 91 |
| | Hypophtalmichtys nobilis | 185 | | 101 | 84 |
| | Monopterus albus (Fluta alba) | 140 | 2 | 102 | 36 |
| | Small fish (species unidentified) | 121 | 1 | 44 | 76 |
| | Hypsibarbus lagleri | 89 | | 19 | 70 |
| | Channa limbata (Channa Orientalis) | 75 | 1 | 68 | 6 |
| | Nemacheilus sp | 53 | | 5 | 48 |
| | Hemibagrus nemurus (Mystus nemurus) | 51 | | 32 | 19 |
| | Macrognathus siamensis | 27 | | 15 | 12 |
| | Oxyeleotris marmorata | 22 | | | 22 |
| | Cirrhinus microlepis | 15 | | | 15 |
| | (Systomus stoiiczkanus) Barbus stoiiczkanus | 13 | | 11 | 2 |
| | Hampala dispar | 12 | | 3 | 9 |
| | Notopterus notopterus | 8 | | | 8 |
| | Mystus mysticetus | 7 | | 5 | 2 |
| | Hypophthalmichthys molitrix | 6 | | 3 | 3 |
| | Ompok krattensis (Ompok bimaculatus) | 5 | | 5 | |
| | Trichogaster trichopterus | 5 | | 2 | 3 |
| | Clupeichthys aesarnensis | 5 | | 2 | 3 |
| | Poropuntius laonensis | 5 | | | 5 |
| | Henicorhynchus caudiguttatus | 4 | | | 4 |
| | Puntioplites falcifer | 3 | | 2 | 1 |
| | Lobocheilus melanotaenia | 3 | | | 3 |
| | Hemisilurus mekongensis | 3 | | 2 | 1 |
| | Cirrhinus cirrhosus | 3 | | 2 | 1 |
| | Barbodes schwanefeldi (Puntius Schw) | 2 | | | 2 |
| | Pangasius macronema | 2 | | | 2 |
| | Pangasius siamensis | 2 | | | 2 |
| | Mastacembelus armatus | 2 | | | 2 |
| | Catlocarpio siamensis | 1 | | | 1 |
| | Thynnichthys thynnoides | 1 | | | 1 |
| | Labeo chrysophekadion | 1 | | | 1 |
| | Osteochilus hasselti | 1 | | | 1 |
| | Botia lecontei | 1 | | | 1 |
| | Bagrichthys macropterus | 1 | | | 1 |
| | Hemibagrus aubentoni | 1 | | | 1 |
| | Bagarius bagarius | 1 | | | 1 |
| | Wallago leeri | 1 | | | 1 |
| | Clupisoma sinensis | 1 | | | 1 |
| | Ompok hypophthalmus | 1 | | 1 | |
| | Pangasius djambal | 1 | | 1 | |
| Crustacean | Potamon sp | 293 | 3 | 271 | 19 |
| | Macrobracium sp | 156 | | 67 | 89 |

**Table 2** (continued)

| Animal | Scientific name (where known) | Total times caught | Unknown | Within ricefield Areas | Outside ricefield areas |
|---|---|---|---|---|---|
| Mollusc | Mollusc species (species unidentified) | 243 | 1 | 123 | 119 |
| | Cipangopaludina chinensis | 193 | 1 | 126 | 66 |
| | Pila sp | 35 | 1 | 24 | 10 |
| | Melanoides tuberculata | 16 | | 12 | 4 |
| Amphibian | Hoplobatrachus rugulosus | 1537 | 14 | 1403 | 120 |
| | Small frog (species unidentified) | 648 | 4 | 604 | 40 |
| | Fejervarya limnocharis | 416 | 4 | 378 | 34 |
| | Tadpoles (species unidentified) | 220 | | 194 | 26 |
| | Occidozyga lima | 46 | | 44 | 2 |
| | Amphibian species unidentified | 24 | | 18 | 4 |
| Reptile | Ptyas Korros | 27 | 2 | 21 | 4 |
| | Lizard unidentified species | 5 | | 4 | 1 |
| | Xenochrophis piscator | 4 | | 4 | |
| | Manouria impressa | 4 | | 3 | |
| | Mabuya multifasciata | 1 | | 1 | |
| | Unidentified reptile | 1 | | | 1 |
| Aquatic insect | Grasshopper (species unidentified) | 80 | 5 | 54 | 21 |
| | Dragon fly | 57 | | 46 | 11 |
| | Aquatic insect (species unidentified) | 22 | 1 | 5 | 16 |
| | Ant eggs | 18 | | 17 | 1 |
| Total identified | | 68 | 18 | 46 | 61 |

resources are important throughout the range of different agro-ecosystem and topographical zones.

It was not possible to estimate the ratio of fish and OAAs to total animal protein but, in the wet season at least, fish are the most frequently consumed form of animal protein and the other aquatic animals are also very important, so one would expect the percentage contribution to be at least as high as that reported by Hortle 2007 (50 %) and most likely higher, particularly in the more aquatic-resource-rich areas. Apart from protein, these resources are an excellent source of calcium, iron and zinc–nutrients scarce in rural Laotian diets (Nurhasan *et al.* 2010).

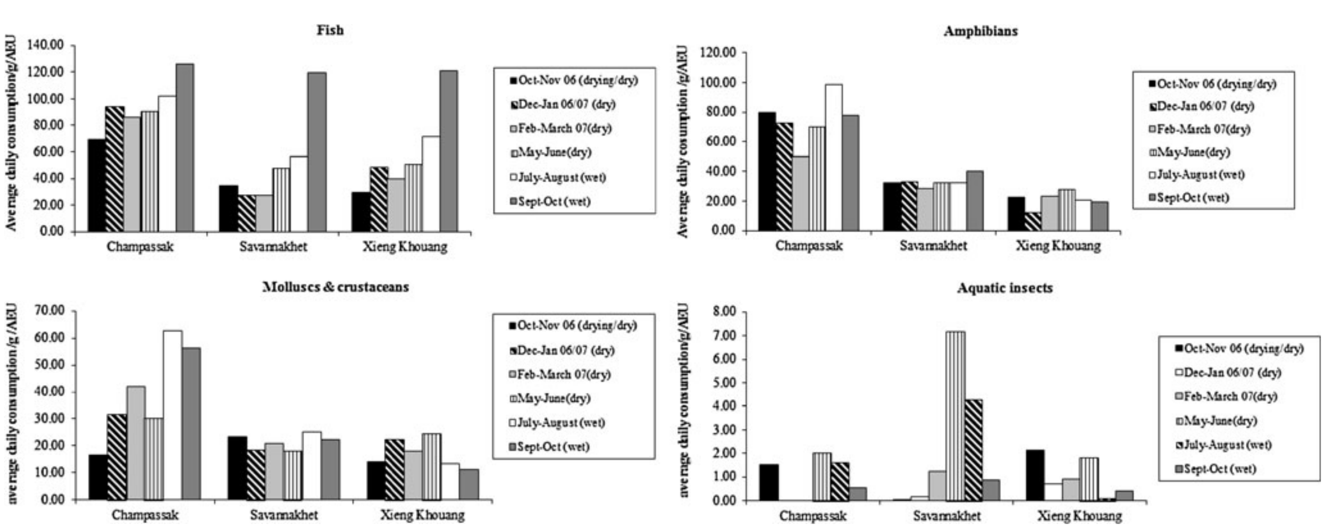
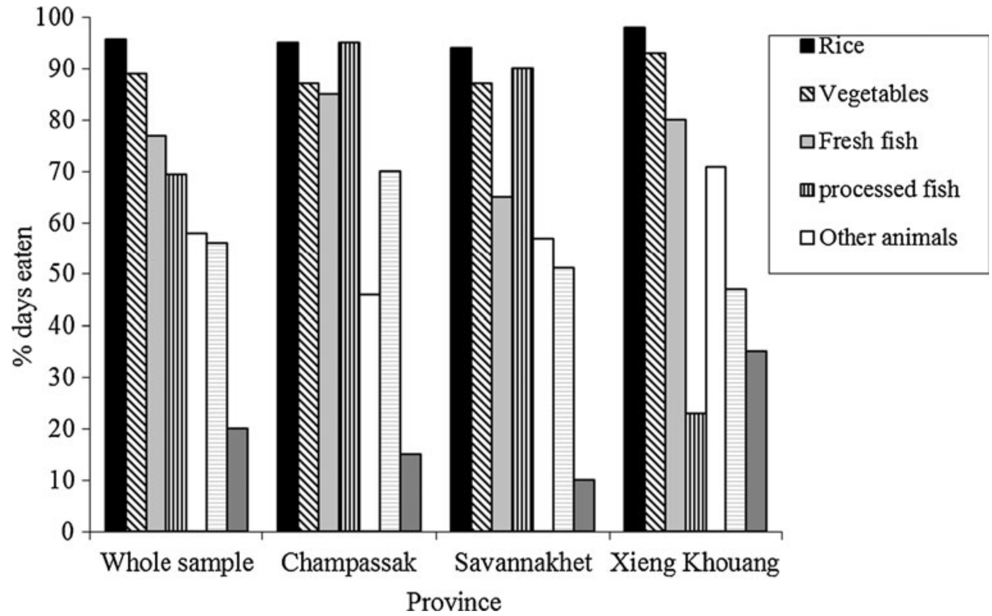



**Fig. 11** Seasonal consumption of fish and OAA's in terms of Fresh Weight Animal Equivalents (FWAE)

Fig. 12 Percentage of days selected foodstuffs were eaten over the 6 days monitored between May 2007 and October 2007 (the *wet* season)




Thus our results confirm that, like wild foods globally, aquatic resources in Lao PDR are a very significant 'hidden harvest' (Scoones *et al.* 1992) that contributes considerably to national food security. It is therefore important to provide accurate statistics on consumption levels for national estimates and that more research focus on their nutritional contribution.

The results also clearly indicate that, throughout the country and in all seasons, more than 90 % of aquatic animals consumed were collected by households themselves, demonstrating the importance of gathering activities for household food security. As with the foods themselves, such activities are frequently missing from mainstream discussions of livelihoods and, consequently, development policy. Rural livelihoods comprise a portfolio of activities and not merely being, for example, a 'farmer' or a 'fisher' or combination of the two. When wild food collection *is* discussed it is often with respect to its importance to the poorest and most vulnerable

**Table 3** Descritive statistics for the frequency (F) of fish and OAA consumption and average (median[Mdn]) quantities consumed (FWAE) when available (g/AEU)

| Animal | Province | Oct–Nov 06 (drying/dry) | | Dec–Jan 06/07 (dry) | | Feb–March 07(dry) | | May–June (dry) | | July–August (wet) | | Sept–Oct (wet) | | Annual FWAE estimates (g/AEU/yr) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | F | Mdn | F | Mdn | F | Mdn | F | Mdn | F | Mdn | F | Mdn | |
| *Padek* (fermented fish) | Champassak | 0.98 | | 0.98 | | 0.98 | | 0.95 | 9.1 | 0.99 | 9.6 | 0.99 | 10.3 | |
| | Savannakhet | 0.93 | | 0.98 | | 0.97 | | 0.84 | 8.2 | 0.94 | 6.7 | 0.93 | 6.2 | |
| | Xieng Khouang | 0.61 | | 0.87 | | 0.93 | | 0.23 | 8.1 | 0.21 | 9.4 | 0.24 | 7.7 | |
| Fish | Champassak | 0.72 | 96.3 | 0.83 | 112.9 | 0.76 | 114.1 | 0.84 | 107.1 | 0.84 | 121.6 | 0.90 | 140.0 | 34616 |
| | Savannakhet | 0.64 | 53.6 | 0.57 | 47.3 | 0.52 | 52.5 | 0.54 | 88.0 | 0.74 | 75.7 | 0.89 | 134.6 | 18994 |
| | Xieng Khouang | 0.69 | 43.1 | 0.72 | 66.8 | 0.68 | 58.5 | 0.69 | 73.2 | 0.81 | 89.2 | 0.86 | 140.3 | 22004 |
| Amphibians | Champassak | 0.63 | 126.4 | 0.60 | 121.1 | 0.46 | 109.7 | 0.63 | 112.7 | 0.66 | 150.2 | 0.41 | 191.0 | 27428 |
| | Savannakhet | 0.50 | 64.3 | 0.55 | 60.6 | 0.49 | 57.3 | 0.41 | 78.7 | 0.51 | 62.9 | 0.31 | 129.2 | 12081 |
| | Xieng Khouang | 0.29 | 76.7 | 0.23 | 54.2 | 0.33 | 70.9 | 0.35 | 79.8 | 0.29 | 69.0 | 0.23 | 83.8 | 7632 |
| Molluscs/crustaceans | Champassak | 0.17 | 96.9 | 0.28 | 114.3 | 0.30 | 140.5 | 0.24 | 123.6 | 0.34 | 182.6 | 0.33 | 168.5 | 14581 |
| | Savannakhet | 0.24 | 98.1 | 0.24 | 76.9 | 0.22 | 95.1 | 0.18 | 98.6 | 0.19 | 129.4 | 0.18 | 125.6 | 7780 |
| | Xieng Khouang | 0.33 | 42.3 | 0.33 | 66.8 | 0.24 | 75.0 | 0.19 | 129.7 | 0.18 | 73.9 | 0.16 | 67.2 | 6271 |
| Aquatic insects | Champassak | 0.08 | 20.4 | 0.00 | . | 0.00 | . | 0.09 | 21.6 | 0.02 | 84.1 | 0.02 | 27.8 | 344 |
| | Savannakhet | 0.01 | 10.6 | 0.01 | 28.0 | 0.04 | 28.0 | 0.24 | 29.3 | 0.17 | 25.2 | 0.04 | 19.9 | 838 |
| | Xieng Khouang | 0.14 | 14.8 | 0.05 | 14.1 | 0.08 | 12.0 | 0.15 | 11.9 | 0.01 | 10.6 | 0.04 | 10.7 | 366 |

households who have less access to land and/or labour. In Lao PDR, wild food collection is carried out by both sexes and all ages, using largely inexpensive equipment, enabling the most vulnerable households to secure food for themselves and their families. However, this should not obscure the fact that it is a mainstream activity important to all households. In addition, the ubiquity of collection reflects a vast amount of local knowledge on wild food species (location, identification, edible parts, harvesting season, methods of collection etc.). As suggested by others (e.g., Bharucha and Pretty 2010), this knowledge is a very important capital resource and efforts should be made to record/conserve it for present and future generations.

Rice-field habitats are far more important than areas outside the rice-fields in terms of total quantities of fish and OAAs collected and (for the OAAs in particular) provide catches far higher than anticipated: 75 % of total fish catch; 93 % of amphibians, 94 % of crustaceans and 63 % of molluscs. Their importance therefore goes far beyond their use for rice production and this multi-functionality needs to be understood and addressed in agricultural policy. Against this backdrop, the current threats to this biodiversity are extremely serious.

At a global level, the recognition that agricultural intensification frequently leads to biodiversity loss has led to increasing attempts to co-ordinate landscape and policy action with maintenance of healthy ecosystems. Scherr and McNeeley (2008) note that while early attempts assumed that this required segregated approaches, more recently a new paradigm has emerged "'eco-agriculture': integrated conservation-agriculture landscapes in which biodiversity is an explicit objective of agriculture and rural development, and the latter are explicitly considered in shaping conservation strategies" (Scherr and McNeeley 2008:477). Bharucha and Pretty (2010) also advocate the integration of conservation, food policy and agricultural policies in order to recognise and preserve the importance of wild foods. Such integration is essential to a country such as Lao PDR where most increases in food supply required over the coming years will need to be produced domestically and potentially on increasingly fragile/marginal lands and where aquatic biodiversity is extremely important for local food supply and livelihoods. At the same time, many of the farming systems in Lao PDR are more ecologically benign production systems than their green revolution (improved seed-fertiliser-pesticide-irrigation) counterparts, and frequently build on, rather than replace, natural ecosystems. Like the local ecological knowledge (LEK) associated with wild aquatic foods, every effort should be made to understand these farming systems and use them as a basis on which to build for the future.

Finally, while this study concentrates on Lao PDR, its conclusions are likely to be relevant to most other countries in the Lower Mekong Region (and perhaps beyond) as similar levels of aquatic biodiversity have been documented here (e.g., Halwart and Bartley 2005) and comparable quantities of fish and OAAs consumed (e.g., Hortle 2007).

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